



Evoltrya OS — Operations Manual



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PART 1 — GETTING STARTED

1.1 Signing in

Open the address your administrator gave you. The sign-in screen asks for **Email** and **Password**, and the button is **Sign in**.

Accounts are created by an administrator under **Settings → Accounts**. There is no self-registration, and the screen says so: "No account yet? Contact an administrator". When your account is created you are handed an initial password in person. The first time you sign in you are sent to **Set your password**, which shows "You are signing in as ... Choose a password to finish setting up your account." The new password must be at least 8 characters. Until you set it, that screen shows no menus and no modules. The only other control on it is **Not you? Log out**.

Three things can go wrong at sign-in, and each says which:

What you see	What it means
Wrong email or password	Check the email for typos, and the capitalisation of your password.
This email has not been confirmed yet	Your password is fine. Open the invitation email and follow the confirmation link, or ask an administrator to send it again.
Too many attempts — try again in a few minutes	Sign-in for this email is paused for a few minutes. Your account is not locked.

If you are signed out while working, the sign-in screen says **Your session has ended** and adds "Anything you had typed on the previous page was not saved." After a period of no activity you are warned before that happens. The warning is headed **You will be signed out shortly** and carries a **Continue working** button, which keeps you exactly where you are.

If you are locked out entirely, an administrator can restore access.

1.2 What you are looking at

After signing in you land on the home page. It carries the company mark, a greeting, and a search box that does not work yet and says so: "Search is not built yet. This box is where it will live — for now it does nothing, so it does not take what you type either." Everything you do is reached from the module bar above it.

If your account holds no module permissions, the home page tells you that your account is signed in, that no module has been granted to it, and to ask an administrator to assign you a role.

What is waiting for you is on **Tools → Reminders**, not on the home page. That page gathers every waiting signal across the modules you can reach.

Immediately after you set your password for the first time you are taken to one of two places. If your account is linked to an employee record, you go to **My profile**. If it is not, you go to a page headed **Your account is ready**, which tells you an administrator will finish setting up your access, and that your own profile, payslips and training will appear once the link to your employee record is made. That is not an error. It means the account exists and the link has not been made yet.

Three pages are yours alone and do not depend on module permissions:

- **My profile** — your own record, payslips and training. Its subtitle reads "Your own record. Only you and HR can see this." If your account is not yet linked to an employee record, it says so and tells you to ask an administrator.
- **My reviews** — performance reviews you conduct, if you conduct any.
- **Notifications** — what the system has raised for you.

1.3 How the navigation is arranged

There is a bar of nine top-level modules across the top:

Purchasing · Logistics · Operation · Sales · Finance · Inventory · HR · Tools · Settings

A module name is a button that opens a menu. It is not itself a link, so clicking a module name never takes you anywhere on its own. The menu under it lists that module's pages.

The menu behaves in six ways that are not obvious from looking at it.

A module opens if any single page under it opens for you. There is no separate "can you enter this module" permission. If you hold only the stocktake permission, the Inventory menu opens and Stocktakes is clickable.

Entries you cannot open are still listed. They are drawn as the entry's own name followed by **Restricted**, and hovering one shows "Requires module access". A module you cannot enter at all is drawn the same way on the top bar. You can always see that a feature exists, whether or not you can use it.

A page can sit under more than one module. Inbound appears under Purchasing, Inventory and Operation. Output appears under Operation and Inventory. Commissions appears under Purchasing and Sales. Freight appears under Logistics and Finance. It is the same page in each case.

Finance is the only module with a third level. Its menu is grouped: **Reports · Journal · Receivables · Payables · Period end · Configuration**, with Overview above the groups.

Some pages are not in any menu. Twenty-one working pages have no menu entry of their own. Every one of them is reached in one click from a page that does have a menu entry, and the door is named wherever the page is described. The full list is in section 3.10. The one people look hardest for is **Field Receiving**, where goods are received against a purchase order: it is a button on the Inbound page, not a menu entry.

Leave has its own tab strip. The HR menu shows one entry, **Leave**. Inside that page a strip of six tabs — **Requests · Balances · Calendar · Annual operations · Leave types · Public holidays** — leads to five more pages; Requests is the Leave page itself. The module menu gives no hint that they are there.

1.4 What a code like PO-2026-0001 means

Every document carries a code with three parts: a prefix saying what kind of document it is, the four-digit year, and a number counted within that year. PO-2026-0001 is the first purchase order of 2026.

The prefixes:

Prefix	Document
PO-	Purchase order

Prefix	Document
SO-	Sales order
QT-	Quotation
INV-	Invoice
CN-	Credit note
SHP-	Shipment
IN-	Inbound batch
OUT-	Output batch
PROC-	Processing run
WO-	Work order
JE-	Journal entry
PMT-	Payment
EXP-	Expense
FA-	Fixed asset
ST-	Stocktake
STMT-	Bank statement
LV-	Leave request
CLM-	Expense claim
MC-	Medical claim
PAY-	Payroll period
ASY-	Assay result
CTR-	Container
CON-	Contract

Two of these count differently, and the difference matters if you ever have to explain a gap. Purchase orders, sales orders, quotations, invoices, credit notes and shipments are numbered without gaps: the system takes the highest number used that year and adds one. Inbound and output batch codes come from a counter that keeps going even if a save is abandoned, so their numbers can skip. A missing **IN-** number is normal. A missing **PO-** number is not.

A contract's number takes its year from the date the contract comes into force, not from the day it was typed in. Everything else takes the year from the document's own date.

You cannot choose a code. The system assigns it when the record is created.

1.5 What "deleted" means here

No document is ever removed from this system. Deleting one leaves the record in place, marked as deleted, together with **who deleted it** and **why**. A reason is required, and the system refuses the deletion without

one.

Deleted records disappear from the lists and detail pages where they used to appear. They are listed on **Settings → Deleted records**, which needs the `data.view_deleted` capability. That page is a register, not a recycle bin: **there is no restore**. Nothing in the system puts a deleted record back.

Deleting a batch is not only a bookkeeping act. For an inbound or output batch, the confirmation says: "This keeps the record and writes a write-off movement for the remaining quantity — it does not erase the batch. Who deleted it and why are both recorded." The stock leaves the books.

Two qualifications:

1. **Some documents cannot be deleted at all.** Invoices, credit notes, journal entries, payments, expenses and shipments refuse deletion outright. So does the stock movement ledger. For these, see 1.6 below and the module sections: the correction is a reversal or a credit note, never a deletion.
2. **Configuration and the rows inside a document are removed outright, not marked.** A public holiday, a storage location's allowed material classes, a line on a draft quotation, a maintenance interval on a machine, a metal on a pricing formula, a step inside a task: removing one of these deletes the row. It is gone, it cannot be undone, and it is not listed on Settings → Deleted records, because there is nothing left to list. The confirmation dialog is the only warning you get.

One wording to be aware of: some delete confirmations still say "(Soft delete: data is kept and recoverable.)" The first half is true. The second half is not something you can act on, as no screen recovers a deleted record.

1.6 Reversal, not correction

Most of Finance, and all of the stock ledger, is append-only. You do not edit a posted document and you do not delete it. You post a second, opposite document.

- A posted **journal entry** is reversed by a reversing entry.
- A posted **payment** is reversed by a reversal.
- An **invoice** can be voided, but only while nothing has been settled against it and nothing has shipped against it.
- A **shipment** cannot be touched once written. The only correction is a **credit note**, and a credit note itself cannot be changed once raised.
- A **stock movement** can never be edited or removed. Stock history only grows.

Read this section before you start entering documents. Wherever a process step in Part 2 has no undo, it says so in those words.

1.7 How permissions affect what you see

Permission in this system answers two different questions, and it answers them in two different ways.

Can you open this page? If not, the page says so: "You do not have access to this module. This is a permission answer, not an empty result — ask an administrator to grant you access to this module." You never get an empty page instead.

Can you see this number? A page you can open may still withhold particular values. The cell reads **Restricted**. It is never blank and never zero. The most common case is money: `data.view_prices` is a separate capability from any module, so a person can open the purchase order list, see every order and every quantity, and see **Restricted** where the amounts are.

The same care runs through empty lists. Where a list is empty because you cannot see its contents, the page says which. On the contract form, for example: "Customers are not listed here because you do not have access to customer records. That is a permission answer, not an empty customer list."

There is one deliberate exception to the pattern. Where a person's name cannot be shown because **this account has no employee record**, that is stated as a fact about the data, not drawn as a permission refusal. A gap in the data and a wall in the permissions are different things and are shown differently.

Every capability has a name. A step that needs `module.finance.edit` needs exactly that string, and that string is what an administrator ticks under **Settings → Roles**. Part 4 lists every capability by name.

1.8 Words used throughout

Batch. A quantity of one material, received or produced as one lot, with its own code. An **inbound batch** (IN-) is material that arrived. An **output batch** (OUT-) is material that came out of processing. Stock is held as batches, not as a single pooled quantity.

Movement. One recorded change to stock: a receipt, a consumption, a sale, a transfer, a write-off, an adjustment. Every stock figure in the system is added up from movements when you open the page. Movements are never edited or deleted.

Assay. A laboratory result saying what a batch actually contains, which metals and at what percentage. It is recorded against a batch and then applied, and applying it restates what the batch is worth.

Journal entry. The accounting record of a transaction, as two or more lines that must add to zero. Journal entries are posted, never edited, and corrected only by reversal.

Ledger. In finance, the accumulated journal entries. In stock, the accumulated movements.

Incoterm. A three-letter trade term (CIF, FOB, DAP and so on) saying who pays for carriage and insurance, and at what point risk passes from seller to buyer.

GST. Singapore goods and services tax. Purchase orders are recorded exclusive of GST; the tax is recorded when the supplier's invoice is entered as an expense.

Base currency. The currency the books are kept in. Foreign-currency documents are converted at a recorded rate, and that rate is stored on the document rather than looked up again later.

Reservation. Stock set aside for a specific sales order line, from a specific batch at a specific location. Reserved stock is still on hand but is no longer available to anything else.

Earmark. An output batch marked as feed for a downstream operation rather than as saleable stock. An earmarked batch cannot be reserved, sold or shipped until the earmark is released.

1.9 Dates

The system almost never fills a date in for you, and where it refuses to, it says why. A process date decides which metal prices a cost allocation uses. An invoice issue date decides which accounting period the entry lands in. A ship date decides which period revenue lands in. An arrival date is the day the goods actually arrived.

Each of these records something that happened in the world, and only the person entering it knows when that was. So blank stays blank. Expect to type a date on almost every document, and expect a refusal that names the field if you do not.

PART 2 — BUSINESS PROCESSES

Real work crosses modules. Receiving one lorry-load touches purchasing, inventory, assay and finance in a single afternoon. This part follows six pieces of work from beginning to end, across whatever modules they pass through. Part 3 is the place to look a single page up.

2.1 Receiving a load into stock

The people involved: whoever weighs and books in the load at the gate; whoever records the condition of the material; whoever enters the laboratory result; whoever prices the batch.

Step 1 — Have a purchase order, or a reason for not having one

Material can be received either against a purchase order line or with no order at all. The second case is legitimate, a free sample or a spot delivery, but the system asks you to say which it was.

- **Page:** Purchasing → Purchase orders → + **New Purchase Order**. See 2.2 for how to raise one.
- **Permission:** `module.purchasing.edit` to raise an order.
- **If not done:** you can still receive, but you must choose a reason under **Source (no purchase order)**. Choosing "other" without typing an explanation is refused: "This reason requires a written explanation."
- **Undo:** an order with nothing received against it can be cancelled with a reason. See 2.2.

Step 2 — Book the load in

There are two doors. **Field Receiving** weighs one material at the scale: kilograms, no price, and it can receive against a purchase order line. **Add Inbound** is the full entry, with unit, unit price and stage.

Field Receiving

- **Page:** Purchasing → **Inbound**, then the **Field Receiving** button at the top right. There is no menu entry for this page.
- **Permission:** the Inbound page needs `module.inbound.view` ; saving needs `module.inbound.edit` .
- **What you enter:** Supplier, Material, **Weighed Quantity (kg)**, **Arrival Date**, and optionally **Declared quantity (optional)** and Notes. The declared quantity is what the supplier said was coming. It is never prefilled from the purchase order, because the order is ours and the declaration is theirs. Left blank it means "not recorded", which is not zero.
- **The button is Save & Get Label.** Saving creates the batch and prints its label. The next screen is headed **Received** and offers **Receive Next**.
- **If the arrival date is missing:** "The arrival date is required — the stock movement records the day the goods actually arrived, and it is never filled in for you. Nothing was saved."
- **Undo:** delete the batch from its own page, giving a reason. That writes a write-off movement for the whole remaining quantity and marks the batch deleted. It is not a way to correct a typo. See the end of this section.

Add Inbound

- **Page:** Purchasing → **Inbound** → + **Add Inbound**.
- **Permission:** `module.inbound.edit` .
- **Against a purchase order:** the form has **Against purchase order** and then **Select an order line...**, showing ordered and remaining quantity per line. If you leave it blank you must give a source reason instead.
- **Undo:** the batch can be edited afterwards (**Edit Inbound**) for most fields. The unit price is not an ordinary field. See step 5.

What happens the moment you save. The batch exists and the material is stock. The receipt movement is written in the same act that creates the batch. Assay and pricing change what the batch is worth; they do not decide whether it is there.

Step 3 — Record the condition on arrival

- **Page:** Inbound → open the batch (**Edit Inbound**), section **Condition on arrival**.
- **Permission:** `module.inbound.edit` .
- **What you record: Safety state**, where you tick everything that is true of the load, and **Chemistry certainty**. A load can be both water-exposed and damaged, and nothing is pre-ticked.
- **The blank is meaningful.** A load with nothing ticked reads "No safety state is recorded for this load. That means NOBODY HAS RECORDED ONE - it does not mean the load is safe." The two chemistry answers are also different from each other: **Mixed** is for material you know to be mixed. Material you could not identify is **Unknown, pending identification**.
- **If not done:** the batch cannot be processed. Committing a processing run that consumes it is refused: "Batch {code} has NO recorded safety state. That means NOBODY HAS RECORDED ONE — it does not mean the load is safe." Record it here, then commit the run again.
- **Undo:** re-open the panel and change the ticks. Saving replaces the whole set with what is ticked, so re-tick everything that is still true.
- **Not every batch answers these questions.** For material that is not battery material the panel says the axes do not apply. Only battery material carries a safety state and a chemistry certainty.

Step 4 — Record the assay result

- **Page:** Inbound → open the batch → + **Record assay**. The page it opens is headed **Record Assay Result**.
- **Permission:** `module.inbound.edit` to record and to apply.
- **Two acts, not one.** Recording the result stores it. **Applying** it is what restates the batch: it updates the recorded metal content and, where the batch is priced from a formula, restates what is payable. The button that does both at once is **Record and apply**; otherwise use **Apply now** on the result.
- **If applied twice:** refused. An assay result can be applied once.
- **If the batch was priced against a contract formula but the terms were never committed:** "{batch} references pricing formula {formula} with no settlement terms recorded at the time." Settlement refuses rather than reading the formula as it stands today.
- **Undo: Unapply**, with a reason. Read the warning before you do: "Unapplying restores the metal content record but does NOT reverse the price change." Reprice explicitly if that is what you want. Unapplying is not a full reversal.

- **A later result supersedes an earlier one.** The superseded result stays on file, marked **Superseded by {code}**.

Step 5 — Price the batch

A batch moves through three pricing states: **unpriced** → **provisional** → **final**.

- **Page:** Inbound → open the batch (**Edit Inbound**), and use the repricing control there.
- **Permission:** `module.inbound.edit` .
- **Why price is not an ordinary field.** The unit price is what is owed to the supplier. Changing it changes the debt, so it goes through its own action and leaves a price-history record. It cannot be typed over in the ordinary way.
- **If the currency has no accepted rate for that date:** the repricing is refused by name and nothing is saved. Enter the rate under **Finance** → **FX Rates** first.
- **Undo:** reprice again. Every price the batch has carried stays on the price history.

What the batch has cost

The batch page shows **Landed cost**: what the batch has cost so far. The purchase price is what is owed to the supplier. Freight and processing cost are capitalised onto the batch separately and never change the purchase price. If the batch has no unit price yet, the page says the landed cost cannot be totalled rather than showing a partial number.

How to undo the whole thing

A receipt stands once it is written. The stock movement written when the batch was created cannot be edited or removed. What you can do is delete the batch with a reason, which writes a write-off movement for whatever is left and marks the batch deleted, with your name and your reason on it. If material was received against the wrong supplier or the wrong material, that is the route, and the reason field is where you say so. There is no restore afterwards.

2.2 Buying, from order to payment

The people involved: whoever negotiates and raises the order; whoever receives against it (2.1); whoever enters the supplier's invoice; whoever pays it.

A note before the first step: approvals are switched off

This system has a two-level purchase approval chain. The chain is built and configured, and it is **not in force**. The approvals page states it: "Approvals are NOT in force. Orders are stamped approved by the system when they are raised; nobody decides them."

So **a purchase order is created already approved**. There is no approval step in the process below, because there is no approval to wait for. Trying to approve one is refused, for the same reason.

Where the state is reported: **Settings** → **Approvals**. That page is read-only. It shows whether approvals are in force, which role approves at level 1, the amount threshold, which role approves at or above it, and what turning the switch would do. It does not carry the switch, and neither does anything else: "This panel is read-

only. There is no screen anywhere in the system for configuring the approval chain." Turning approvals on is a change made directly to the database.

If approvals are ever switched on, an order is then raised as a draft and must be approved before goods can be received against it or a deposit released. Turning the switch on leaves existing orders receivable, and turning it off never retroactively approves anything.

Step 1 — Raise the purchase order

- **Page:** Purchasing → Purchase orders → + **New Purchase Order**.
- **Permission:** `module.purchasing.edit`.
- **Choose the kind first.** An order is either **Materials** or **Equipment**, never both. The rule is enforced when you submit, so choosing before you type saves the typing. Mixing them is refused: "One order is either all material or all equipment."
- **What is required:** an **Order date**, a supplier, a currency with an accepted FX rate, and at least one line. Refusals, by name: "An order date is required — it decides which FX rate values the order"; "Add at least one line".
- **Equipment orders are different.** Each line is one machine, and the machine's asset card must already exist: "Pick a machine registered under Finance → Assets." A purchase order line never creates a machine. An equipment order is also never received into stock: a machine arriving creates no batch, has no assay and does not enter a location, so the order carries no receive action at all.
- **GST.** Amounts on a purchase order are exclusive of GST. An order is a commitment, not a tax point. Input GST is recorded later, when the supplier's invoice is entered as an expense, using that supplier's default tax code.
- **Payment terms** can be attached from a template (**Purchasing → Payment terms**), setting instalments with their triggers and due dates. Percentages across the instalments cannot exceed 100.
- **Undo: Amend** on the order, with a reason. Quantity, price and dates can be amended. Supplier and currency cannot: changing either makes it a different deal, so raise a new order instead. Every change is recorded. Changing the order date re-derives the FX rate for that date, and if no rate is on file the amendment is refused rather than a rate being guessed. Amending a closed order is refused until you reopen it: "This order is closed. Reopen it first."

Step 2 — Receive against it

Covered in full in 2.1. The relevant fact here is that receiving is what turns an order into something you owe against.

- **If the order does not exist or was cancelled:** you cannot receive against it. Receive with a stated source reason instead, or raise the order.

Step 3 — Record the supplier's invoice as an expense

- **Page:** Finance → Expenses → + **New Expense**.
- **Permission:** `module.finance.edit`.
- **What it does:** posts to the ledger. This is where input GST is recorded. Enter the amount **net** of GST and pick a tax code. The tax is computed on top: a claimable code posts it to input tax, a blocked code posts it into the expense itself. The supplier's default code is offered, and where there is none the form refuses

to guess: "No default is available, so a code must be picked here." TX, ZP, EP, BL and OP all behave differently on the return.

- **Non-resident payees.** If the payee is recorded as non-resident, the form asks whether tax must be withheld and will not accept a blank. Where withholding genuinely does not apply, choose **Not subject to withholding** rather than leaving the field empty. An expense that attracts withholding cannot be recorded as already paid; record it unpaid and withhold at the payment.
- **Capital purchases.** Ticking **Capital expenditure (fixed asset)** posts to fixed assets and creates the register entry in the same transaction, so the asset cannot exist without the payable behind it.
- **If the equipment line has already been expensed:** refused. "One line is expensed once." If the price changed, amend the order line instead.
- **Undo: there is no undo.** An expense cannot be edited or deleted. The correction is a **reversal**, which posts an opposite entry and leaves both on the ledger.

Step 4 — Pay

- **Page:** Finance → Payments → + **Record Payment**.
- **Permission:** `module.finance.edit` .
- **What is required:** a payment date, a direction, a counterparty of the right kind for that direction, an amount, a currency and its rate, and the documents the payment settles.
- **Currency must match.** "{document} is a {currency} document — settle it with a {currency} payment (got {other})."
- **You cannot over-allocate.** Allocating more than a document's open amount is refused, and so is allocating more in total than the payment is for.
- **You cannot allocate to another party's document.** "Document {code} belongs to a different counterparty."
- **Deposits.** Where a deposit was paid on the order, releasing it against an invoice is a separate act on the invoice. Releasing moves that money from prepayments onto the invoice; no cash moves. The deposit is measured in base currency at the rate it was paid at, the release amount is in the invoice's currency, and the server converts.
- **Undo: there is no undo.** A payment cannot be edited or deleted. The correction is a **reversal**.

Step 5 — Close the order

- **Page:** the order's own page, **Close order**.
- **Permission:** `module.purchasing.edit` .
- **Refused if:** the order is cancelled, or already closed.
- **Notes are required where a deposit is unresolved:** "This order has {amount} of unapplied prepayment — a note explaining how it is resolved is required to close."
- **Undo: Reopen**, with a reason. Reopening returns the order to the status it held before it was closed.

Cancelling an order

Cancelling is narrow.

- **Refused once anything has been received:** "Cannot cancel: {n} inbound batch(es) are linked to this order".
- **Refused once a deposit has been released:** "Purchase order {code} already has a deposit released against it." Cancelling would leave that release pointing at a void order, so reverse the release first.
- **A reason is required.**
- **There is no undo.** A cancelled order is closed for good and cannot be reopened. Who cancelled it and why are both recorded.

If goods have already arrived, the order cannot be cancelled at all. Close it instead, with a note.

2.3 Processing a batch into output

The people involved: whoever plans the work; whoever releases the plan; whoever records what actually ran; whoever settles the cost.

A **work order** is the plan. A **processing run** is what happened. They are separate records, and a run does not have to have a plan behind it.

Step 1 — Write the work order (optional)

- **Page:** Operation → Work orders → + **New work order**.
- **Permission:** `module.processing.edit`.
- **What it is:** a plan of what to process, how much, and when. What actually happened lives on the processing runs, and the progress figures on the work order are read from those runs rather than stored.
- **Planned inputs are by material, not by batch.** When a plan is written the batches often do not exist yet, and which batch to use is a decision made on the day.
- **Expected outputs are optional**, and every expectation row must say where its number came from: **Planner estimate**, **Seeded from industry experience (low confidence)** or **Calibrated against real production**. There is no default. An empty expectation table means nobody estimated, which is not the same as expecting zero.
- **It saves as a draft.** Releasing it is a separate step.
- **Undo: Amend plan**, with a reason. A planned quantity cannot be reduced below what linked runs have already consumed.

Step 2 — Release the work order

- **Page:** the work order's own page, **Release**.
- **Permission:** `module.processing.edit`.
- **If not done:** a run cannot be worked against it. "Work order {code} is {status} — only a RELEASED order can be worked against." A draft has not been agreed yet; a closed or cancelled one has already ended.
- **Refused if not a draft:** only a draft can be released.
- **Undo:** there is no un-release. A released order is closed or cancelled, never returned to draft. Cancelling is refused once any run has been worked against it: "Work order {code} already has {n} processing run(s)

against it, so it cannot be cancelled — close it instead." Cancelling would say the work never happened, and the material really did move.

Step 3 — Record the processing run

- **Page:** Operation → Processing runs → + **Add Processing Run**.
- **Permission:** `module.processing.edit` .
- **What you choose:**
 - **Work order (optional)** — only released orders are listed. Inputs are not prefilled from the plan: a plan names materials and a run names batches.
 - **Operation** — the operation decides what the machine accepts. Some operations produce no new batch at all: the same batch goes in and comes out with its state changed.
 - **Process Date** — required. It decides the movement date and which metal prices the allocation uses. The Save button is disabled while it is blank.
 - **Cost allocation basis** — prefilled from the company default in Finance settings, and changeable here. It decides each output batch's reported gross margin, so it is a choice, never a default.
 - **Inputs (consume inbound)** — batches and quantities. The same batch cannot be added twice.
 - **Outputs (produced goods)** — material, quantity, purity. Total output cannot exceed total input.
- **If the input batch has no condition recorded:** refused, with the route out named. See 2.1 step 3.
- **If the input carries a safety state that may not be fed:** "Batch {code} carries safety states that may not be fed: {states}." Every one of them has to be cleared. A load that is discharged and water-exposed is still water-exposed; discharging it does not cancel the water.
- **If the operation does not accept what the batch is carrying:** the refusal says so, and it is not the same answer as "may not be fed". Another operation may well accept the batch: charged material goes through Deep discharge first, and packs that cannot be discharged go to the Battery processing line.
- **If the machine is not yet acquired, or has been disposed of:** refused by name.
- **Undo: Delete Processing Run** on the run's page, which asks "Roll this processing run back?" and confirms with **Roll back this run**. A reason is required. Rolling back reverses the run: the consumed inputs are restored to their batches, the output batches are voided, and reversing movements are written to the stock ledger. It cannot itself be undone. A reversed run stays listed against its work order, because it really was worked, but its consumption stops counting anywhere.

Step 4 — Allocate the cost

This is a separate, later step. Committing a run does not do it.

- **Page:** the run's own page, **Cost Allocation**.
- **Permission:** `module.processing.edit` . Allocating cost is a processing right, not a finance one: the run decides how its own cost is spread.
- **What it does:** puts cost onto the output batches. Committing a run produces the batches; allocating is what gives them a unit cost.
- **If not done:** the run sits committed and unallocated indefinitely, and the batches carry no cost. The page says: "This run has cost entries but has never been allocated." The ledger carries them and no batch does. Month-end close refuses while any run is in this state.

- **If the cost entries changed after the last allocation:** the page says "Allocation is out of date — cost entries changed after the last allocation". Re-running it posts the difference to inventory for the in-stock share, to cost of goods sold for the sold share, and to inventory adjustment for the written-off share, dated today.
- **If allocating by metal value and no output batch has a priced metal:** "No priced metal content on any output batch." Record assay results, add a price for at least one contained metal under Metal Prices, or allocate by weight instead.
- **Undo:** re-allocate. Allocation records themselves cannot be edited.

Step 5 — Decide what the output batch is for

An output batch carries two independent answers: how much of it has been sold, and what it is for. The second is set by hand.

- **Page:** Operation → Output → open the batch (**Edit Output**), section **Purpose**.
- **Permission:** `module.processing.edit`. Promising a batch to the plant is a processing decision, not a sales one, so a person with sales rights alone cannot change it.
- **What it means:** whether this batch is saleable stock, or is already earmarked as feed for a downstream operation. This is a separate axis from the sales state: a batch can be both in stock and already promised to the plant.
- **If earmarked:** selling, reserving and shipping are all refused. That is not a statement that the material may not be sold. Release it back to saleable stock and it sells.
- **Undo:** change the purpose back. Both directions need processing edit rights.

Step 6 — Record the safety state of the output batch

Self-produced material faces the same gate as bought material.

- **Page:** Operation → Output → open the batch (**Edit Output**), section **Safety state**.
- **Permission:** `module.output.edit`.
- **If not done:** "No safety state has been recorded. That means NOBODY HAS LOOKED, not that the batch is safe." The batch cannot be fed into any operation until someone records one here.

2.4 Selling, from quotation to cash

Material leaves this company by one of two routes. They are separate processes, not two ways of doing the same thing.

The order flow starts from a quotation, produces a sales order, reserves specific stock, invoices, then ships. It posts to the ledger at invoicing and again at shipment.

The direct sale records a sale straight off an output batch. It moves stock and posts at the moment of sale. Invoicing afterwards is a separate, later act that gathers posted sales onto a document and posts nothing.

2.4.1 The order flow

The order is invoiced before it is shipped. Most people describe a sale the other way round, so this is the first thing to fix in your head. The system enforces it line by line, and the invoicing panel states it on screen: "Order flow requires INVOICE BEFORE SHIPMENT."

Step 1 — Raise the quotation

- **Page:** Sales → Quotations → **New quotation.**
- **Permission:** `module.sales.edit` .
- **What it is:** the document before the commitment. A quotation touches no stock and no ledger. Its one special power is converting into a sales order, copying the quote exactly.
- **The customer does not have to be onboarded.** A name and a country are enough, so a quotation can go out before the customer is set up.
- **Both dates are required and neither is defaulted.** The validity date is a commitment, and a defaulted one would never expire on the day it should.
- **Saving creates a draft.** Issuing is a separate step, and issuing is what sends it to the customer.
- **Undo:** a quotation stays editable after issue, because negotiating is what it is for. Changing it after issue puts a banner on the page: "This quotation has been changed since it was last issued — the copy the customer holds no longer matches." Re-issue it to send the current version. Each issue appends a version, and older versions are kept.

Step 2 — Issue it

- **Page:** the quotation, **Issue PDF.**
- **Permission:** `module.sales.edit` .
- **If there are no lines:** "Add at least one line first." An issued quotation with no lines is a blank page to the customer.
- **The system does not know whether the customer received it.** There is no "sent" flag, because the system cannot see what the counterparty received.

Step 3 — Convert it to a sales order, or record the decline

- **Page:** the quotation, **Convert to sales order** or **Record decline.**
- **Permission:** `module.sales.edit` .
- **If the quotation was never issued:** "Issue this quotation first." A draft has not been offered to anyone yet.
- **If it expired:** "This quotation expired on {date}, so it cannot be converted." Change the validity date and issue a new version, so that the customer is looking at a price that is still on offer.
- **If it was declined, or already converted:** refused by name, naming the order it became.
- **The order date is the day the customer accepted, not the quotation date.** It decides the order number's year and the FX period, so it is never filled in for you.
- **What conversion does:** creates a draft sales order copying the quotation exactly — same lines, quantities, prices, price provenance, currency and FX rate, with nothing re-derived. The quotation then freezes, so that the offer and the order it produced cannot drift apart.
- **Declining requires a reason.** Three months later nobody remembers why a quotation did not close.

- **Undo: there is no undo of a conversion.** The quotation is frozen from that moment: "Quotation {code} became a sales order and can no longer be changed." If the order was wrong, cancel the order and raise a new quotation.

A sales order can also be raised directly at Sales → Orders → **New order**, without a quotation.

Step 4 — Confirm the order

- **Page:** the order's own page, **Confirm order**.
- **Permission:** `module.sales.edit` .
- **What it does:** freezes customer, currency, FX rate, order date and all lines. Notes stay editable.
- **If the customer is on credit hold:** "Customer {code} is on credit hold, so this order cannot be confirmed." Lift the hold under Customers, or leave the order as a draft.
- **If the order has no lines:** refused. An order with nothing on it cannot be confirmed.
- **Undo:** a confirmed order can be **cancelled**. It cannot be returned to draft.

Step 5 — Reserve the stock

Shipping consumes a reservation, so this step is not optional. Only a reservation says which batch and which location the goods come from.

- **Page:** the order's own page, **Reserved stock** section, **Reserve**.
- **Permission:** `module.sales.edit` .
- **If the order is not confirmed:** "Order {code} is a {status}, so stock cannot be reserved against it." A draft is not yet a promise; confirm the order first.
- **Only output batches can be reserved.** Sales draw from output batches, and incoming material has to be processed first.
- **If the batch is earmarked for processing:** "Batch {code} is earmarked as {purpose}, so it is not saleable stock and can be neither reserved nor shipped." That is not a statement that the material may not be sold. Release the earmark on the output batch page, or use a different batch.
- **If you try to reserve more than the line is for:** "Cannot reserve {qty} — this line is for {line} and {already} is already spoken for (shipped plus reserved)." Reserving more would promise the same line twice.
- **If the batch does not hold that much:** "Cannot reserve {qty} — only {available} is available in that batch and location." Reserving does not create stock; it sets aside what is already there.
- **What reserving does:** moves that quantity into the committed bucket for the line. Sales and processing cannot touch it until it is released here, or shipped.
- **Undo: Release**, with a reason. A reason is required because taking back a promise is exactly what nobody can reconstruct later. Releasing returns the stock to available. A partial release cancels the reservation and records a new one for the remainder, so each row stays a true record of what was promised. The system never releases a reservation for you.

Step 6 — Invoice

- **Page:** the order's own page, **Invoicing** section, **Create invoice ({n} line(s))**. This is on the sales order, not on the Finance → Invoices page.

- **Permission:** seeing the section needs `module.finance.view` ; raising the invoice needs `module.finance.edit` , because an invoice is a posting document.
- **If the order is still a draft:** "Only a confirmed order can be invoiced."
- **The issue date decides the posting period and is required.** It is never defaulted.
- **What it posts:** AR against contract liability, at the order's rate. Shipping requires this invoice. The credit limit is checked here, because invoicing is where the exposure is created.
- **Per line.** You can invoice part of an order, and then ship only that part.
- **Undo: void the invoice,** but only while it is still voidable. Voiding is refused once anything has been settled against it, and refused once anything has shipped against it. After that the correction is a **credit note**.

Step 7 — Ship

- **Page:** the order's own page, **Ship**. You pick which reservation is being shipped.
- **Permission:** `module.sales.edit` .
- **A ship date is required:** "The ship date is required — it is the day the goods physically left, and it decides which period the revenue lands in. Nothing was saved."
- **If the line is not on a live issued invoice:** "Line {n} of order {code} is not on a live posted invoice." Order flow is invoice-before-shipment: raise the invoice first. Nothing is shipped in the meantime.
- **If the line is not reserved:** "No active reservation on this line." Reserve the stock first.
- **If you ship more than is reserved:** "Cannot ship {qty} — that reservation is for {reserved}." Reserve more first, or ship what is reserved.
- **If the order is not in a shippable state:** refused. Only a confirmed or partially shipped order can ship.
- **What it does:** writes a shipment (`SHP-`), moves the stock out, releases the contract liability into revenue at the rate stored on the invoice, and moves the order to **Partially shipped** or **Shipped**. Shipping creates no new receivable: the debt was recognised when the invoice was raised, and that invoice stays the one and only AR row for the order.
- **Undo: there is no undo.** A shipment cannot be edited or deleted, and the invoice behind it can no longer be voided.
- **One line on the confirmation dialog is wrong.** It says corrections would go through a credit note "which does not exist yet". Credit notes do exist. Finance → Credit notes is the register, and one is raised from the invoice.

Partially shipped is a dead end

Read this before you ship, not after.

Once an order reaches **Partially shipped** it has no legal move to any other status. Not back to confirmed, not on to shipped except by shipping the rest, not to cancelled, not to closed.

The way to correct a partially shipped order is a **credit note**. It is raised on the invoice's own page: **Finance → Invoices → the invoice → Raise a credit note**, which needs `module.finance.edit` . The Credit notes page in the menu is the register, not the place a credit note is created. A credit note is immutable from the moment it is created. It has no status, it cannot be edited, and it cannot be deleted. Read it back before you save it.

Step 8 — Take the cash

- **Page:** Finance → Receivables, or Finance → Payments → **+ Record Payment**.
- **Permission:** `module.finance.edit` .
- **What it does:** settles the invoice.
- **Undo:** there is no undo. A payment is reversed, not deleted.

2.4.2 The direct sale

Where material is sold straight off an output batch with no order behind it.

Step 1 — Record the sale

- **Page:** Operation → Output → open the batch (**Edit Output**), section **Record Sale**.
- **Permission:** `module.output.edit` .
- **What is required:** quantity, unit price, currency, exchange rate, sale date, and normally a customer. The sale date sets the FX rate used, the inventory movement date, and the period both journals post to, so it is never filled in for you.
- **Pricing** can be a manual price, a spot price at market, or a formula. Choosing a formula and pressing **Compute price** fills it in and shows the working. Editing the figure afterwards reverts it to manual.
- **If no customer is chosen:** the form warns rather than refusing. The sale is recorded, but it is not credit-checked and it does not count towards anyone's exposure. It belongs to nobody until a customer is attached to it later, and it cannot be invoiced to a customer before that.
- **Credit control:** if the customer is on hold, or the sale would take them past their limit, the sale is refused. The panel shows the limit, the outstanding and the headroom before you press the button.
- **If the batch is earmarked for processing:** refused, as for reservation.
- **What it does:** moves the stock out and posts both the revenue and the cost of goods sold.
- **Undo:** there is no undo. The sale record cannot be edited or deleted, and the stock movement it wrote cannot be removed. A credit note is not the route here. Credit notes apply to order-flow invoices only. For a direct sale, the correction is to void the invoice or reverse the receipt.

Step 2 — Invoice the sale

- **Page:** Finance → Invoices → **+ New Invoice**.
- **Permission:** `module.finance.view` to open the page, `module.finance.edit` to save it.
- **What it does:** gathers posted sales that are not yet on a live invoice onto one invoice for a chosen customer. This kind of invoice does **not** post a journal entry. The ledger entries were already made when the sale was recorded.
- **Sales with no customer are listed** and marked as such. They can be invoiced to the customer you choose.
- **Undo:** void the invoice, subject to the same limits as above. A credit note cannot be raised against this kind of invoice.

2.5 Stocktake

The people involved: whoever counts on the floor; whoever reviews the differences and posts.

Step 1 — Open the stocktake

- **Page:** Inventory → Stocktakes → + **New Stocktake**.
- **Permission:** `module.stocktakes.edit` .
- **What it does:** opens a count (`ST-`) with status **Open**, listing the batches to be counted.
- **Undo:** **Cancel Stocktake**, with a reason. The counted lines are kept for the record but will never be posted. Who cancelled it and why are both recorded.

Step 2 — Count

- **Page:** the stocktake's own page, or the batch page while a count is running. The batch page shows a banner: "Stocktake {code} in progress — record this batch's count."
- **Permission:** `module.stocktakes.edit` .
- **What you enter:** a counted quantity per batch, and optional notes. The page shows **Current**, **Book**, **Counted** and **Delta** side by side, and summarises how many batches have been counted and how many of them differ.
- **Undo:** **Re-count** re-opens a line. Nothing has left the ledger yet.

Step 3 — Review and post

- **Page:** the stocktake → **Review & Post**.
 - **Permission:** `module.stocktakes.edit` .
 - **What posting does:** writes one adjustment movement per differing batch and sets its stock to the counted quantity. If there are no differences the page says so, and posting changes nothing.
 - **If the stocktake is not open:** refused. "Stocktake is not open (status: {status})."
 - **If a batch on the count has since been deleted:** "Batch {code} has been deleted."
 - **Undo:** **there is no undo**. A posted stocktake is final and has no transition out. The adjustment movements it wrote cannot be removed. To correct a wrong count, open a new stocktake and count again.
-

2.6 Month-end close

The people involved: finance, with HR for the payroll steps.

Closing a period locks it. The work is to get every open item settled first, and the **Month-end** page lists those items as steps you can walk down.

- **Page:** Finance → Month-end.
- **Permission:** `module.finance.view` to read it. Each step's own page has its own permission.

The page shows ten steps, each marked **done**, **outstanding**, **blocked** or **n/a**, each linking to the page that clears it:

#	Step	Where it is cleared
1	Board rates entered for every foreign-posting day	Finance → FX Rates
2	Payroll posted	HR → Payroll
3	Employees paid	Finance → Salary payments
4	CPF remitted	Finance → Salary payments
5	Deductions remitted	Finance → Salary payments
6	Processing accruals settled	Finance → Cost settlement
7	Batch costs match cost entries	Operation → Processing runs
8	Depreciation posted	Finance → Fixed assets
9	Revaluation run	Finance → Revaluation
10	Period locked	Finance → Close

Three of these report themselves as blocked rather than merely outstanding: employees cannot be marked paid before the payroll is posted; a currency with no mid rate on file for the date has to have one entered under FX first; and the revaluation has to be run before the step that depends on it.

CPF falls due after this close. It is due by the 14th of the following month and its journal lands in that month, so locking this month does not block it.

Locking the period

- **Page:** Finance → Close.
- **Permission:** `module.finance.edit`.
- **Refused, in this order:**
 - the date given is not a month end;
 - that period is already closed;
 - depreciation has not been run for the period: "Depreciation of {amount} is still outstanding — the period cannot be locked until it is posted." Locking first would make the charge unreachable without reopening the period;
 - a committed processing run still has no cost allocation;
 - the trial balance does not balance.
- **Undo: Reopen**, on the same page, with a reason. The period's entries become editable again, the close record is kept, and the lock date moves back. Reopening the same close twice is refused by name. Reopening a month does not reopen a closed financial year: an entry dated inside a closed year is still refused until the year itself is reopened.

Closing the financial year

Also on the Close page, and separate from the monthly lock. It has its own preview and its own checks: the final month closed, trial balance level, revaluation level and depreciation level. Draft payroll periods and open accruals are flagged as warnings and do not block it.

Closing the year takes every P&L account to zero and posts the net result to 3100. The year then refuses new entries until it is deliberately reopened.

Undo: Reopen year, with a reason. The closing entry is reversed, the audit trail is kept, and the year accepts entries again.

Filing GST

Separate from the month-end close, and quarterly.

- **Page:** Finance → GST.
 - **Permission:** `module.finance.edit` .
 - **The return must tie before you file:** "THE RETURN DOES NOT TIE — do not file it until the difference is explained." The page names the two checks that can fail: documents against statute, and documents against ledger.
 - **A quarter cannot be filed while its books are still open:** "This quarter ends on {end} but the books are only closed up to {locked}." A filed return whose entries can still change is not a record of anything.
 - **The filing itself happens elsewhere.** Filing happens on the IRAS portal. This system records what was filed, when, and by whom; it submits nothing.
 - **Undo: there is no undo.** A filed return is a record of what was filed and is never edited. The correction is **Raise a correction**, which creates a new return pointing at the one it corrects, leaving the original exactly as it was filed.
-

PART 3 — MODULE REFERENCE

Nine modules sit on the top bar.

Where a state is **final**, nothing moves the document out of it.

Where a page has no menu entry, the entry point is given. Section 3.10 lists them all in one place.

Seven of the nine modules open with a menu entry called Overview. Tools and Settings have none. The page itself is headed with the module's name rather than the word Overview, so "Purchasing → Overview" opens a page headed **Purchasing**. Six of the seven carry only the facts that span several of the module's pages and that no single page can state on its own; the pages themselves are in the menu above. Inventory's is older and different in shape; see 3.6.

3.1 Purchasing

What you buy, who you buy it from, and what has arrived against it.

Page	Menu	What it does
Overview	yes	What is still coming in — orders confirmed or part-received, how many are past their expected delivery date, and how much buying runs under a contract.
Purchase orders	yes	The order register. Raise, amend, close, reopen and cancel orders.
Receiving discrepancies	yes	Where what arrived does not match what was ordered: short, over, declared against weighed, a different material than ordered, or an assay outside tolerance. Each is measured against a threshold set under receiving settings.
Payment terms	yes	Instalment templates that can be applied to an order: share, trigger and due date per instalment.
Suppliers	yes	Supplier master data, compliance certificates and attachments.
Contracts	yes	The contract register, and the only place a contract is created.
Licences	yes	This company's own licences, not a supplier's: number, kind, issuing body, validity and the approved storage limit. A supplier's licences live on that supplier's own compliance record.
Commissions	yes	Commission agreements with agents. Also under Sales.
Inbound	yes	Received batches. Also under Inventory and Operation; see 3.6.
Field Receiving	no — from Inbound	Weigh and book in one material at the scale, optionally against a purchase order line.

The purchase order

Code **PO-**. Two kinds, chosen when it is raised and never mixed: **Materials** and **Equipment**.

States: Draft · Confirmed · Receiving · Closed · Cancelled.

From	To	By
Draft	Confirmed	Approving it. See the note below on why you will not see this in practice.
Confirmed / Receiving	Closed	Close order , with notes if a deposit is unresolved
Closed	back to its previous state	Reopen , with a reason
any live state	Cancelled	Cancel order , with a reason, and only while nothing has been received and no deposit released

Final: Cancelled. A cancelled order is closed for good and cannot be reopened. Closed is not final: it reopens.

Approval status is a second, separate marker on the order: Pending · Approved · Rejected. Both Approved and Rejected are final, and neither can be changed once set. Because approvals are switched off, every order is created with approval status **Approved** and status **Confirmed**. In practice you will never see a draft purchase order.

Deleting: marked, not removed.

The contract

Code `CON-`. Created at Purchasing → Contracts → **New contract**.

Creating a contract needs edit rights on whichever side it is with: `module.suppliers.edit` for a contract with a supplier, `module.customers.edit` for one with a customer. Reading the register needs `module.suppliers.view`, so a person who holds only customer access cannot open the register at all.

What you record: who the contract is with, a **Kind** (Supply, Offtake, Framework, Service, Other), a **Title**, an **In force from** date, an optional **In force until** (blank means no fixed end, and means exactly that), optional headline terms (signed date, currency, incoterm, payment terms in days, document reference, notes) and a **Status**.

A contract is with a supplier or with a customer, never both. Picking a supplier makes it a buy-side contract; picking a customer makes it sell-side. Where the same company is both to you, that is two contracts, because they are two agreements.

Status is chosen at creation and is permanent. The two choices are **Active** and **Draft**:

- **Active** — purchase and sales orders can be linked to this contract, and a linked order copies its terms.
- **Draft** — still being negotiated. The contract is recorded and readable, but no order can be linked to it, and anything raised against it is refused.

The form states plainly that the choice cannot be revisited: nothing in the system moves a contract from one status to another, so whichever you pick is what it stays. Pick Draft only if you are content for the contract to stay unusable.

The other three values a contract can hold — Suspended, Expired, Terminated — appear in the register but cannot be chosen at creation and cannot be reached afterwards.

After creation, the only thing you can do with a contract is read it. It cannot be edited, amended, moved to another status or deleted. Record it only when you are content with what it says.

There is also no screen yet for linking an order to a contract. A contract can be recorded and read, and the register reports against it, but no page attaches a purchase order or a sales order to one.

Undo: there is none. A contract recorded wrongly stays in the register as recorded.

Refusals when creating one:

- "A contract is with exactly one party — either a supplier or a customer, not both and not neither. Nothing was saved."
- "The contract needs a title, so that a person reading the register can tell what the agreement is. Nothing was saved."
- "The date the contract takes effect is required." Without it the register cannot say whether the agreement was in force when an order was raised.
- "The end of the term is earlier than its start, so nothing was saved."
- "Payment terms must be a whole number of days between 0 and 365. Nothing was saved."
- "Recording a contract needs permission to edit whichever side it is with."

What a contract is for. A purchase order is not a contract. A long-term supply agreement is: it is the relationship a document is raised under. Contract terms — grade specification, insurance obligations, volume commitments — live on the contract, and a document raised under one copies the terms in force at that moment. Later edits to the contract do not reach it. The register also reports grade specifications not met, measured against the specification the document copied when it was linked.

Read the coverage figures before the breach figures. Nothing forces a document to carry a contract, and a spot purchase does not run under an agreement in the first place. So "no contract breached" can also mean "nobody linked anything".

Payment term templates

A template holds instalments: a label, a share, a trigger event and a due date offset. Applying one to an order writes its schedule onto that order. Percentages across instalments cannot exceed 100. A template with a fixed-amount instalment must state the currency that amount is in.

Editing a template replaces all its lines. Templates are read only when applied, so changing one does not reach orders that already used it.

Removing an instalment line removes the row outright. It is not marked as deleted and it is not listed on Settings → Deleted records.

3.2 Logistics

Getting goods from one place to another, and the paperwork that has to travel with them.

Page	Menu	What it does
Overview	yes	Where every container has got to, counted by its latest milestone; how many shipments are not yet attached to a container.
Forwarders	yes	Freight forwarders, their rate quotes and contacts.
Lanes and document checklists	yes	Routes, and which documents each route requires.

Page	Menu	What it does
Containers	yes	Containers (CTR-), their milestones, their ETA and their documents.
Freight	yes	Freight invoices. Also under Finance.

Logistics has no edit capability. There is `module.logistics.view` and nothing else. Writing to logistics data is gated on `module.purchasing.edit`. On the role screen the Edit box for Logistics is drawn as a dash, with a note that the module has no separate edit permission and its View permission is the whole of it.

Freight

A freight invoice is recorded at Finance → Freight → **+ Record freight**, on a page headed **Record a freight invoice**. It needs `module.finance.edit`.

Freight is capitalised. It goes into the cost of the material it carried. The page states the trade-off that comes with that: a wrong apportionment now sits inside inventory instead of showing on the P&L.

You choose the basis on which the freight is spread across batches, by weight or by value. The two diverge most when it matters most, on a light expensive batch travelling with heavy cheap material. Pick the one the invoice actually supports; where the forwarder itemised per batch, use that.

Undo: Reverse this freight document. A reversed document has no reverse button, because reversing it again is refused by name.

3.3 Operation

The plant: what was planned, what ran, and what came out.

Page	Menu	What it does
Overview	yes	Plan against what actually happened, and how much output is still mid-process.
Work orders	yes	The plan register (WO-).
Processing runs	yes	What actually ran (PROC-), with inputs, outputs, loss and cost.
Work in progress	yes	Output batches earmarked as feed and still holding quantity — what is waiting, how much, and for which operation.
Shift handover	yes	What one shift hands to the next.
Inbound	yes	Also under Purchasing and Inventory.
Output	yes	Also under Inventory.

Work order

Code WO- . **States:** Draft · Released · Closed · Cancelled.

From	To	By
Draft	Released	Release
Draft	Cancelled	Cancel order , with a reason
Released	Closed	Close , with a reason

From	To	By
Released	Cancelled	Cancel order , refused once any run has been worked against it

Final: Closed and Cancelled. There is no un-release and no un-close.

Closing a plan short of its quantity is allowed and recorded. A rule forcing you to shrink the plan before closing it would erase the variance instead.

Amending a released plan is allowed with a reason, but a planned quantity cannot go below what the linked runs have already consumed.

A work order has no deleted marker. It cannot be deleted at all.

Processing run

Code **PROC-** . **States:** Committed · Reversed.

A run is created committed. The only move is **Delete Processing Run**, with a reason, which sets it Reversed, restores the consumed inputs, voids the output batches and writes reversing movements. **Reversed is final.**

A reversed run stays listed against its work order, because it really was worked against that order, but its consumption no longer counts anywhere.

Loss is recorded by category. Water and volatiles leave but the metal stays. Dust and spillage take the metal with them. Residue sent for disposal is not a loss at all. Collapsed into one number, recovery could never be right. Categorised amounts need not add up to the run's loss total, but may not exceed it.

Cost allocation is a separate act, described in 2.3 step 4.

Shift handover

Records what one shift hands to the next: the date, the shift, who handed over, who took over, and the content. The incoming person **acknowledges** it, and until they do the state is named rather than blank: "Unacknowledged means THE NEXT SHIFT HAS NOT SAID THEY READ THIS." It does not mean the field is empty.

A handover does not record what the shift processed, because a processing run carries only a date and no time of day, so no run can be attributed to a shift. Nor does it record incidents: those belong on the workplace incident and near-miss register, which is not yet built. Equipment tick boxes point at downtime already recorded elsewhere rather than restating it.

3.4 Sales

Offering, committing, and getting paid.

Page	Menu	What it does
Overview	yes	Where the work is sitting across quote → order → shipment → invoice, and credit exposure with the customers who have no limit set.
Quotations	yes	The quotation register (QT-).

Page	Menu	What it does
Orders	yes	The sales order register (SO-), with reservation, invoicing and shipping on each order's page.
Customers	yes	Customer master data, credit limits and holds.
Commissions	yes	Commission agreements. Also under Purchasing.
Counterparty overlap	no — from Customers	Companies that are both a customer and a supplier.

Quotation

Code **QT-** . **States:** Draft · Issued · Declined · Converted.

From	To	By
Draft	Issued	Issue PDF
Issued	Declined	Record decline , with a reason
Issued	Converted	Convert to sales order

Final: Converted and Declined. A converted quotation is frozen: "Quotation {code} became a sales order and can no longer be changed." The offer and the order it produced must not drift apart.

A quotation stays editable after issue, and re-issuing appends a version rather than replacing one. Older versions are kept, because the copy the customer holds is a specific version.

Deleting: marked, not removed.

Sales order

Code **SO-** . **States:** Draft · Confirmed · Partially shipped · Shipped · Closed · Cancelled.

The permitted moves, in full:

From	To
Draft	Confirmed, or Cancelled
Confirmed	Cancelled (and to Partially shipped / Shipped by shipping)
Partially shipped	nothing
Shipped	Closed
Closed	final
Cancelled	final

Partially shipped is a dead end. It has no legal move out. Not back, not forward except by shipping the remainder, not to cancelled. **The way to correct a partially shipped order is a credit note**, raised on the invoice's own page. That is the undo slot for this state, and there is no other.

Final: Closed, Cancelled, and in practice Partially shipped.

The order page lists the moves available from wherever it is, rather than listing what is forbidden. The database holds the same table and always wins. Where nothing is left, the page says the order is in a final

state with no further transitions.

Deleting: marked, not removed.

Shipment

Code `SHP-` . **No status at all, and no way to change it.** A shipment is written by shipping an order and nothing else writes one. It cannot be edited and it cannot be deleted.

Undo: there is none. The correction is a credit note.

Customers

Customer master data: legal name, country, contacts, credit limit and credit hold. A customer's status is free text with no state machine behind it. It is an attribute of the record, not a document lifecycle.

Credit control happens in two places. On the order flow it is applied when the invoice is raised, because invoicing is where the exposure is created. On a direct sale it is applied when the sale is recorded. A customer on hold blocks both.

A sale recorded with no customer belongs to nobody. It can be attached to a customer afterwards on the receivables page, and that attachment is **one-way**: it cannot be changed to a different customer or undone.

3.5 Finance

The ledger, and everything that posts to it. This is the largest module and the only one with a third level in its menu: **Reports · Journal · Receivables · Payables · Period end · Configuration**, with Overview above them.

Everything here needs `module.finance.view` to read. Anything that posts needs `module.finance.edit` .

Overview

The state of the books: which day the books are locked to, the ledger against the sub-ledgers for receivables, payables, raw stock and finished stock, and the net position. Where a figure cannot be reached the page says so rather than printing a zero, because answering anyway would give a confident 0.00.

Reports

Page	What it does
Trial Balance	Every account and its balance.
P&L	Profit and loss for a period.
Balance Sheet	Assets, liabilities and equity as at a date.
Cash flow	Cash movement for a period.
Cash forecast	What is expected to come in and go out, week by week over thirteen weeks, in the currency it actually moves in. Amounts with no date are listed separately and are not in any week.
Price exposure	Material bought on a floating price against material sold on a fixed price. The purchase side is not modelled, and the page states that rather than showing a zero.

Page	What it does
Batch Gross Margin	Revenue per output batch against its allocated processing cost. Needs <code>data.view_prices</code> and either finance or processing access.

The margin page carries a distinction that catches people. Its cost figure is the run's current allocated unit cost multiplied by the quantity sold. The general ledger posts cost of goods sold once, at the moment of sale, and never restates it. The two figures are different and both are correct.

Journal

Journal lists entries. **New Entry** raises one by hand.

Journal entry, code `JE-`. **States:** Posted · Reversed. The only move is a reversal. **Reversed is final.** A journal entry cannot be edited and cannot be deleted.

Receivables

Page	What it does
Receivables	What customers owe, by document, with ageing.
Invoices	The invoice register (<code>INV-</code>), and + New Invoice for gathering posted direct sales onto a document.
Credit notes	The credit note register (<code>CN-</code>). A credit note is raised from the invoice it credits, not from here.

Invoice, code `INV-`. Two kinds. An **order** invoice is raised from a sales order and posts to the ledger. A **sale** invoice gathers already-posted direct sales and posts nothing.

States: Issued · Void. The only move is **void**, and voiding is refused once anything has been settled against the invoice or anything has shipped against it. **Void is final.** An invoice cannot be edited and cannot be deleted.

Credit note, code `CN-`. **No status column and no transitions. It is immutable from the moment it is created**, and cannot be edited or deleted.

A credit note reduces what the customer owes on that one invoice, after the invoice can no longer be voided. It is not a refund: no money moves, and it cannot take the invoice below zero.

Credit notes apply to order-flow invoices only. For an invoice that gathered direct sales, the correction is still to void the invoice.

The credit note date is required and never defaulted. It decides which accounting period the reversal lands in, and a defaulted date could never hit a locked period, so leaving it blank would slip through where filling it in correctly does not.

Payables

Page	What it does
Payables	What is owed, by document, with ageing.
Payments	The payment register (<code>PMT-</code>), and + Record Payment .
Expenses	The expense register (<code>EXP-</code>), and + New Expense .

Page	What it does
Expense claims	Staff expense claims (CLM-) raised in HR and turned into payables here.
Fixed assets	The asset register (FA-), depreciation and disposal.
Freight	Freight invoices. Also under Logistics.

Payment, code PMT- . **States:** Posted · Reversed. The only move is a reversal. **Reversed is final.** It cannot be edited or deleted.

Expense, code EXP- . Its marker is a payment status: Unpaid or Paid. The correction is a reversal. It cannot be edited or deleted.

Fixed asset, code FA- . **States:** Active · Disposed. Moves: put in service, record acceptance, dispose. **Disposed is final.** An asset card has no deleted marker.

An asset card can carry maintenance intervals and, where the purchase order had retention terms, a retention clock. **Retention is never paid automatically.** When it falls due, confirm that the machine gave no trouble during the retention period, then state how much is released and how much is withheld.

Removing a maintenance interval removes the row outright, and it means something specific: that this machine is no longer watched for this kind of work. That is not the same as leaving it monitored but off the dashboard.

Period end

Page	What it does
Month-end	The ten closing steps, each linking to the page that clears it. See 2.6.
Salary payments	Pay employees, remit CPF, remit deductions.
Cost settlement	Turn processing accruals into real invoices and post the variance.
Revaluation	Restate foreign-currency balances at the closing rate.
Cost variance	Estimated against actual processing costs, by cost type across months.
Close	Lock a period; reopen a locked one with a reason.
GST	The quarterly return and its filing record.
Withholding tax	Tax withheld from payments to non-residents, and what is owed to IRAS.
Monthly pack	The month in one document: statements, ageing, reconciliation, and what it cannot see.
Bank	Bank reconciliation.

Cost settlement distinguishes two things on one page: actual costs waiting to be remitted, and estimates waiting to be relieved with the real invoice. Relieving posts the variance, and the invoice date decides which month it lands in — relieving a July accrual with an August date puts the variance in the wrong month.

Revaluation refuses without a closing mid rate for every foreign currency in play. Re-running the same date posts nothing.

Close. A period close records who closed it and when. Reopening asks for a reason and warns that the period's entries become editable again. Closing the financial year is a separate act with its own preview:

every P&L account is taken to zero, the net result posts to 3100, and the year then refuses new entries until it is deliberately reopened.

GST period. States: Open · Filed. **Filed is final** — a filed return is a record of what was filed and is never edited. The correction is a new return that points at the one it corrects. A GST period has no deleted marker.

Monthly pack. A stored pack means the month was already closed when it was produced. Open months can be previewed and exported but are never stored.

Bank — reconciliation. **Bank statement**, code `STMT-`. **States:** Open · Reconciled, and this one is reversible: reconciling and unreconciling both exist.

Two sub-pages have no menu entry, both reached from the Bank page: **Statements** and **Import Statement**. Importing maps CSV columns to fields, and the mapping can be saved for next time. The reconciliation identity is printed on the page: closing balance, plus items on your books the bank has not shown, plus items the bank shows that are not on your books, equals the ledger balance.

Transfer between own accounts is on the Bank page. Both amounts are entered exactly as the bank reported them, in their own currencies, so both statements can reconcile.

Configuration

Page	What it does
Settings	Company-wide finance settings, including the default cost allocation basis and the period lock date.
Company	Company details as they print on documents, and GST registration.
FX Rates	Exchange rates by date and currency.

FX Rates has one sub-page with no menu entry, reached from the FX page: **Enter a week**, a grid for filling gaps. Cells that already hold a rate are read-only there. To change one, open it from the link in the cell and give a reason, which keeps the previous number on record.

Rate history is immutable. A rate is corrected by recording a new one with a reason, not by overwriting.

3.6 Inventory

What is on hand, where it is, and what it is worth.

Page	Menu	What it does
Overview	yes	Headed Inventory . Material balance across all processing runs, and current stock by material, backed by the movement ledger. This overview is a stock listing rather than the set of cross-page facts the other module overviews carry.
Storage locations	yes	Locations, their zones, and which material classes each one accepts.
Stocktakes	yes	Physical counts (<code>ST-</code>).
Materials	yes	The material dictionary.
Reports	yes	Four read-only reports; see below.
Inbound	yes	Received batches (<code>IN-</code>). Also under Purchasing and Operation.
Output	yes	Produced batches (<code>OUT-</code>). Also under Operation.

Reports

The Reports page is a card wall. All four sub-pages are reached from it and none is in the menu. Nothing on them is stored: every figure is derived from the movement ledger when you open the page.

Report	What it shows
Stock snapshot	What is on hand, by material, location and status, with ageing.
Class violations	Stock sitting where its material class is not allowed.
Safety stock	Monitored materials and how they stand against their thresholds.
Movement ledger	Every stock movement, filterable by date, material and batch.

The snapshot values stock at landed cost, which is purchase price plus freight plus capitalised processing cost. That is the same definition used by write-off, stocktake and the general ledger reconciliation. Without `data.view_prices` the value columns read **Restricted**, which is not zero: the quantities are complete and only the money is withheld.

Stock with no location recorded is shown as its own group. That is a normal state rather than missing data — the goods are real, and a transfer can assign a location at any time.

Inbound batch

Code `IN-`. Its marker is a **pricing status**: Unpriced · Provisional · Final. It also carries a **Stage**: To Process · Processing · Processed.

There is no state machine in the usual sense. Pricing status follows from what has been recorded, and repricing moves it. See 2.1.

Deleting: marked, with a reason, and it writes a write-off movement for the remaining quantity.

A known fault in the exported file. Download the inbound list as CSV and the stage column holds the stored value, which is in Chinese (待加工 / 加工中 / 已加工完), not the English shown on screen. The screen and the file will not read the same for that one column. Nothing else is affected, and the stored data itself is unchanged. The output batch export does not have this fault.

Output batch

Code `OUT-`. It carries **two independent markers**, and confusing them is the most common mistake on this page.

State — how much has been sold. In stock · Partially sold · Sold out. **This is written by the system**, not chosen. A person picks the starting value once, when creating the batch, and after that selling and shipping maintain it. Note what Sold out does and does not mean: it means sold, not empty. A batch consumed by a downstream process also reaches zero remaining, and does not get this state.

Purpose — what the batch is for. Saleable stock, or earmarked as feed for a downstream operation. This is chosen by hand and needs `module.processing.edit`. An earmarked batch cannot be reserved, sold or shipped.

Deleting: marked, with a reason, and it writes a write-off movement.

Stocktake

Code **ST-** . **States:** Open · Posted · Cancelled.

From	To	By
Open	Posted	Confirm & Post
Open	Cancelled	Cancel Stocktake , with a reason

Both Posted and Cancelled are final. A posted stocktake has no move out, and the adjustment movements it wrote cannot be removed. See 2.5.

Storage locations

A location has a code, a name, a zone and a set of allowed material classes. A location with no allowed classes shows **not configured**, which the page distinguishes from allowing nothing: zero rows means nobody has decided yet.

Putting stock in a location that does not accept its class is refused, and the refusal names three ways out: pick another location, add the class to that location, or correct the material's classification.

Removing an allowed class removes the row outright. It is not marked as deleted and it is not listed on Settings → Deleted records.

Materials

The material dictionary: name, category, unit, classification and form. Material status is free text with no state machine. It is master data, not a document.

The **form** decides saleability. A material whose form may not be sold under the law cannot be quoted or sold. Where the form was never set the refusal is a different one, and says so: that is not a statement that the material may not be sold, and the way out is to set the form under Materials.

3.7 HR

People, pay, time and performance.

Page	Menu	What it does
Overview	yes	Headcount in service, the attendance and payroll cycle, and total monthly fixed gross.
Employees	yes	The employee register, employment history and documents.
Departments	yes	Departments and their heads.
Attendance	yes	Monthly sheets of overtime and unpaid leave.
Payroll	yes	Payroll periods (PAY-) and their posting.
Leave	yes	Leave requests (LV-), plus five tabbed sub-pages.
Claims	yes	Medical claims (MC-).
Training	yes	Training records.
Reviews	yes	Performance reviews, plus two sub-pages.

Page	Menu	What it does
KPI	yes	KPI scorecards.
KPI scoring	no — from KPI	Entering scores. Needs <code>module.hr.edit</code> .
Review cycles	no — from Reviews	Opening a review cycle.
Rating scale	no — from Reviews	The rating scale reviews are scored against.
Organisation chart	yes	Reporting lines.

Everything here needs `module.hr.view` . Pay figures additionally need `data.view_pay` , identity and work pass numbers need `data.view_identity` , and review content needs `data.view_reviews` .

Leave

The Leave page carries its own tab strip — **Requests · Balances · Calendar · Annual operations · Leave types · Public holidays** — and it is the only way to reach the five sub-pages behind it. Requests is the Leave page itself.

Leave request, code `LV-` . **States:** Pending · Approved · Rejected · Cancelled.

From	To	By
Pending	Approved or Rejected	Approve / Reject
Pending or Approved	Cancelled	Cancel leave

Approved, Rejected and Cancelled are all final as far as the request is concerned. Cancelling releases the days back to the grants they came from, and the ledger keeps both entries.

Days are counted by the system, not typed: weekends and public holidays are excluded, and the count is computed by the database. Where Monday-to-Friday counting is wrong, on a six-day or shift schedule, there is an explicit exception path that asks for the days and a reason. An exception can exceed a leave type's standard days but can never exceed an accrued annual leave balance, because that balance is a real entitlement rather than a policy.

Refusals when submitting a request:

- "Not enough leave: {available} days available, {requested} requested."
- "You will have {n} days accrued by then, and you asked for {m}." Annual leave is earned monthly, and the message names the date from which there will be enough.
- "Annual leave cannot be taken during probation." It keeps accruing month by month from the start date, and whatever has accrued becomes bookable on confirmation.
- "These dates overlap request {code}."
- "A medical certificate is needed: {n} days already taken this year against a {m}-day allowance."

Annual operations is where the year's entitlement is granted and where last year's unused balance is carried forward. Both are safe to run twice: a second run is refused rather than doubling anyone up.

Leave types and **Public holidays** are configuration. Day counts, certificate thresholds and active flags are data, changed here without a release. Public holidays are added year by year as they are gazetted, and lunar

dates are not computed — they come from the official announcement. **Removing a holiday removes the row outright**, and it is not listed on Settings → Deleted records.

Medical claims

Code MC- . **Stored states:** Submitted · Approved · Rejected · Paid.

From	To	By
Submitted	Approved or Rejected	the decision
Approved	Paid	raising the expense and paying it

Paid and Rejected are final.

The list screen shows more than four, because it refines Approved by how far the money has got:

Submitted · Approved · Rejected · Awaiting expense · Expense raised · Part paid · Paid. Those extra three are read from the linked expense and its settlements, not stored on the claim. The claim itself is in one of the four states above.

A claim is settled by turning it into a payable. **Raise expense** creates an unpaid expense on staff welfare and medical, settled through the usual payment run. That step needs finance permission: HR approves the claim, and finance turns it into a payable.

Refused if the claim exceeds the remaining entitlement, if it is not approved yet, or if an expense has already been raised for it.

Expense claims

Code CLM- . **States:** Submitted · Withdrawn · Approved · Rejected. A claim is decided or withdrawn, and all three end states are final. An expense claim has no deleted marker.

Payroll

Payroll period, code PAY- . **States:** Draft · Posted. **This one is reversible:** **Post to ledger** and **Unpost** both exist, and unposting asks for a reason. Unposting reverses the journal entry and returns the period to draft.

Figures are prefilled from last month, with a warning to check every row against the provider report before saving.

Attendance

Attendance period. States: Open · Complete.

From	To	By
Open	Complete	Mark complete
Complete	Open	Reopen , with a reason, and only while that month's payroll is not posted

The sheet is what is reported to the payroll provider each month: overtime hours and unpaid leave. Nothing on it is calculated into pay; the provider does that.

Marking complete freezes the sheet and lets that month's payroll be posted. It is refused while anyone is unanswered: "{n} line(s) still have nobody's answer." Recording zero is an answer; leaving a line blank is not.

Once payroll is posted the sheet can no longer be reopened. Unpost the payroll first.

Performance reviews

States: Draft · Self-assessment · Submitted · Approved · Acknowledged · Void.

A review is created by opening a review cycle, or raised as a probation review from the employee record. A probation review cannot be raised without a probation end date on file, because that date is what the confirmation decision turns on and what the reminders count down to.

Goals carry a target and a unit, and both are set together. A goal without a unit can never take a number later: neither the self-assessment nor the reviewer can add the unit afterwards.

Finalising a self-assessment locks it, and an empty one cannot be corrected without the reviewer reopening it.

Review content needs `data.view_reviews`. Without it, a person sees their own and no others, and the page says so rather than showing an empty list.

3.8 Tools

Small things that do not belong to one module.

Page	Menu	What it does
Tasks	yes	The task board.
Calendar	yes	Everything dated, on one month.
Unit converter	yes	Mass, metal grade, and wet-to-dry weight.
Reminders	yes	What is waiting and what is falling due, across every module.
Pricing	yes	A hub of three pricing pages.

Calendar is read-only and open to any signed-in person, but each item obeys its own module's permission. It gathers everything dated that you can already see elsewhere onto one month. It is a place to look rather than a place to edit: click an item to go to the page that owns it. Where a source could not be read, the page says the month is incomplete rather than showing it as empty.

Unit converter is open to any signed-in person and reads no business data. Three conversions, each showing its formula and where its numbers come from. The wet-to-dry conversion calls the same database function the settlement path calls, so this tool and an invoice cannot disagree.

Reminders gathers waiting signals from every module, and an item appears only if your permissions reach its own module. It separates three answers carefully: a real zero, where you can see those items and right now they hold nothing; a permission wall, where the account cannot answer whether they hold anything; and nothing visible at all.

Pricing is a card wall with three sub-pages, none of them in the menu:

Card	What it does
Pricing formulas	Agreed payable percentages and charges per counterparty.

Card	What it does
Calculator	Value a batch from its assay and current metal prices.
Metal prices	Every price recorded, newest first.

Under Metal prices there is a fourth page, one level further down: **Bulk entry**, headed **Daily Metal Prices**, for entering a day's prices in one go. Reading metal prices is open; writing them needs `module.pricing.edit`.

Removing a metal from a pricing formula removes the row outright.

Tasks

States: To do · In progress · Done. A task is either **Personal** or **Team**.

A personal task is private, and no permission overrides that. The page says: "This is a personal task. Only you can see it and change it." Nobody else can read it: not a colleague, not a manager, not an administrator. It is a boundary of the system rather than a permission somebody could be granted, and no screen offers to grant it.

Make this a team task promotes a personal task. Once other people have worked on it the promotion cannot be reversed, because turning it personal again would leave them unable to read what they worked on.

Promotion needs your login to be linked to an employee record. If it is not, the page says to ask HR to link the account.

Steps. A task can carry steps, and a step can carry one level of sub-steps. Steps are reordered and renamed in place. **Deleting a step removes the row outright**, as with the other child rows and configuration described in 1.5: it is not recorded on Settings → Deleted records and nothing puts it back. Read the confirmation before you accept it. A step that has sub-steps is refused rather than removed with its children.

Deleting a task marks it, as with any other document.

3.9 Settings

Administration.

Page	Menu	What it does
Accounts	yes	System accounts, the roles they hold, and the employee record each is linked to. Create an account.
Roles	yes	Roles and their capabilities.
Permission reference	yes	The full capability catalogue, and who holds each one.
Dictionaries	yes	Reference lists used by materials and receiving.
Bulk import	yes	Load rows from a CSV file.
Approvals	yes	The purchase approval chain, read-only.
Deleted records	yes	What was deleted, by whom, and why.

Accounts, Roles, Permission reference and Approvals all need `action.manage_permissions`. Deleted records needs `data.view_deleted`. Bulk import needs `action.bulk_import`. Dictionaries needs either materials or inbound access.

Accounts. An account is created here, with exactly one role: an account with no role will not be created. The account can sign in as soon as it exists, and the initial password is handed to the person directly, who is then required to change it at first sign-in. An account can be linked to one employee record, and an employee already linked elsewhere is not offered.

Removing a role from a person asks for a reason, and it is recorded.

Bulk import loads materials, counterparties, employees, departments and storage locations from CSV. It is the only action in the system that inserts hundreds of rows at once. Each template's third line lists the accepted values for every restricted column.

Deleted records is read-only, and reading is all it offers. No screen in the system puts a deleted record back.

3.10 Pages with no menu entry, and where to find them

Twenty-one working pages are not listed in any menu. Each is one click from a page that is.

Page	Where the door is
Field Receiving	Inbound → Field Receiving
Import Bank Statement	Finance → Bank → Import Statement
Bank Statements	Finance → Bank → Statements
Enter a week of rates	Finance → FX Rates → Enter a week
Leave balances	HR → Leave → Balances tab
Leave calendar	HR → Leave → Calendar tab
Annual leave operations	HR → Leave → Annual operations tab
Public holidays	HR → Leave → Public holidays tab
Leave types	HR → Leave → Leave types tab
KPI scoring	HR → KPI → KPI scoring
Review cycles	HR → Reviews → Review cycles
Rating scale	HR → Reviews → Rating scale
Movement ledger	Inventory → Reports → Movement ledger
Safety stock	Inventory → Reports → Safety stock
Stock snapshot	Inventory → Reports → Stock snapshot
Class violations	Inventory → Reports → Class violations
Counterparty overlap	Sales → Customers → Counterparty overlap
Price Calculator	Tools → Pricing → Calculator
Pricing Formulas	Tools → Pricing → Pricing formulas

Page	Where the door is
Metal Prices	Tools → Pricing → Metal prices
Daily Metal Prices	Tools → Pricing → Metal prices → Bulk entry

PART 4 — ROLE QUICK REFERENCE

4.1 Roles are data

The roles described below are the ones this company uses today. They were fitted to how the company is organised now, and they are not built into the system.

Permissions are composed one capability at a time. A role is a name with a set of capabilities attached, and nothing anywhere in the system tests a role by name. Every gate asks whether the signed-in person holds a particular capability, and that resolves through whichever roles they hold. To every gate in the system, a role created this afternoon works exactly like one that has been there since the beginning.

New roles can be created, existing ones renamed or re-granted, and people moved between them, as the company grows. None of it needs a developer, a code change, a migration or a deployment. The roles page says the same thing in one line: "Roles and their grants are data — changing them is an edit here, not a release."

4.2 How to create a role

Everything below is gated on one capability, `action.manage_permissions`.

1. Sign in as somebody holding `action.manage_permissions`.
2. Go to **Settings** → **Roles** and choose **Add role**. The page notes the order of work: "Save the role first, then set its permissions on the next screen."
3. Fill in the role's **Code**, its **Name (EN)** and **Name (ZH)**, optional descriptions in both languages, and a sort order. The code is a stable identifier such as `warehouse`, and it is permanent: "It cannot be changed later", and afterwards "The code is fixed after creation — policies and grants identify the role by it."
4. Save. **The role now exists and grants nothing**. It has zero capabilities until you give it some.
5. Open the role and tick capabilities in the permission matrix, then **Save permissions**.
6. Go to **Settings** → **Accounts** and give the role to a person.

Two rules the matrix enforces while you tick:

- **Edit requires View**. "A role that can change records but not read them cannot save at all, because writes read the row back. Ticking Edit ticks View; unticking View unticks Edit." If edit is granted without view the save is refused by name rather than quietly corrected.
- **A module with no separate edit capability shows a dash, not a box**. Logistics is the only one: "This module has no separate edit permission — its View permission is the whole of it."

Renaming, reordering and deactivating a role are all done on the role's own page. Retiring one is done from the role list, and it warns first how many people will lose it.

4.3 The one thing nobody can do

The administrator role is protected. It cannot be deleted, cannot be deactivated, cannot be soft-deleted, and its protected flag cannot be cleared. All four attempts are refused by name.

Alongside that, the system refuses to be left with no administrator at all. Removing the last administrator role from the last person holding it is refused: "This would leave the system with no administrator, so it was refused. Grant the administrator role to somebody else first, then remove it here."

The protected flag is not something the interface can set, so a second protected role cannot be made. Everything you create through the interface is an ordinary role.

4.4 What the capabilities are

There are 39 capabilities, in three kinds. Thirty-eight of them can be granted. The thirty-ninth is held by no role and offered by no screen.

Module capabilities say which part of the system you reach. Most come in a pair, view and edit.

Capability	What it covers
module.customers.view / .edit	Customer master data
module.finance.view / .edit	Ledger, receivables, payables, payments
module.hr.view / .edit	Employees, payroll and training
module.inbound.view / .edit	Inbound batches and receiving
module.inventory.view / .edit	Inventory and material balance
module.logistics.view	Forwarders, lanes, rate quotes, containers and shipping documents — view only; there is no edit capability
module.materials.view / .edit	The material dictionary
module.output.view / .edit	Output batches and sales
module.pricing.view / .edit	Pricing formulas, calculator, metal prices
module.processing.view / .edit	Processing runs and traceability
module.purchasing.view / .edit	Purchase orders and payment schedules
module.sales.view / .edit	Sales orders — create, confirm, cancel, issue
module.stocktakes.view / .edit	Physical counts and adjustments
module.suppliers.view / .edit	Supplier master data
module.tasks.view / .edit	The task board

Data capabilities say which values you see on pages you can already open. They cut across every module, and this is where the **Restricted** cells come from.

Capability	What it reveals
data.view_prices	Unit prices, pricing formulas, costs and margins
data.view_pay	Salary, CPF and payroll figures
data.view_identity	Identity numbers and work pass numbers
data.view_reviews	Ratings, written conclusions, self-assessments and goal results in performance reviews
data.view_sales	Quantity, unit price, amount, customer and date of sales made from output batches

Capability	What it reveals
<code>data.view_banking</code>	Company bank account name, number, SWIFT and bank address as printed on invoices
<code>data.view_deleted</code>	The deleted-records register, across every module. Audit-natured, not a day-to-day permission.

Action capabilities are two operations that stand apart from any module.

Capability	What it allows
<code>action.manage_permissions</code>	Create roles and change who holds what
<code>action.bulk_import</code>	Load materials, counterparties, employees, departments and storage locations from a CSV file. The only action that inserts hundreds of rows at once.

The thirty-ninth capability is `module.tasks.view_all`, which would let its holder read other people's **personal** tasks. **No role holds it, and no screen can grant it.** Personal tasks are private from everybody, including an administrator.

4.5 The roles in use today

Every role defined in the system is listed here. Roles marked **held** have at least one person in them today; the rest are defined and waiting for one. A role with no holder still works — it grants exactly what is listed against it the moment somebody is given it.

System Administrator — `admin` · held · protected

"Full access to everything, including who can see what."

Holds every capability except `module.tasks.view_all`, which nobody holds. It is the only protected role: it cannot be deleted, deactivated or stripped of its flag. One other role, Commercial & People, also holds `action.manage_permissions` and can therefore administer accounts and roles.

General Manager — `gm` · held

"Sees the whole business, including costs and margins, but cannot change who has access."

Edit rights across purchasing, sales, customers, suppliers, materials, inbound, output, inventory, stocktakes, processing, pricing, finance, HR and tasks. View on logistics. Sees prices, sales, banking and review content. **Does not hold** `action.manage_permissions`, so it cannot change roles or grants, and it does not see the deleted-records register.

Commercial & People — `cco` · held

"Customers, sales and everything about people, plus system administration: accounts, roles and permissions. Reads every other module without changing it."

Edit rights on sales, HR, materials and tasks. Read on everything else. Holds `action.manage_permissions` and `action.bulk_import`, and every data capability. This is the second role that can administer accounts and roles.

CFO — `cfo` · held

"Approves purchase orders at or above the approval threshold. Deliberately cannot raise purchase orders."

Five capabilities: view on finance, purchasing and logistics, plus pay and price visibility. It holds no edit capability at all, so the role that approves a large order cannot also raise one.

Approvals are switched off, so this role approves nothing today. See 2.2.

Finance — **finance** · held

"Ledger, payables, receivables, invoicing and payments, with full cost visibility."

Edit on finance, purchasing, suppliers, customers, materials, inbound, output, inventory, pricing and tasks. View on logistics. Sees banking, pay, prices and sales. No HR, no processing, no stocktakes.

Operations Supervisor — **operations** · held

"Runs processing, inventory and stocktakes: quantities, yields and recovery, never prices."

Edit on processing, inbound, output, inventory, stocktakes, materials and tasks. View on logistics. **Holds no data capabilities at all**, so every money column reads **Restricted**. That covers unit prices, costs and margins alike.

Warehouse & Field — **warehouse** · held

"Receiving, output and stock counts on the floor; no commercial data."

Edit on inbound, output, inventory, stocktakes and tasks. View on logistics. No prices, no sales figures, no purchasing, no processing.

Procurement — **procurement** · defined, nobody holds it today

"Negotiates and raises purchase orders; sees prices but cannot pay anyone."

Edit on purchasing, suppliers, materials, pricing, inbound and tasks. View on inventory and logistics. Sees prices. **No finance capability**, so it cannot record an expense or a payment.

Sales — **sales** · defined, nobody holds it today

"Customers, output batches and sales; invoicing is handled by finance."

Edit on sales, customers, output, inventory, pricing and tasks. View on materials and logistics. Sees prices and sales figures. **No finance capability**. A holder of this role can raise and confirm an order and reserve stock against it, but cannot raise the invoice. Because the order flow is invoice-before-shipment, that also means the role cannot ship. See 2.4.1.

Human Resources — **hr** · defined, nobody holds it today

"Employee records, payroll and training, including pay and identity data."

Edit on HR and tasks. Sees pay, identity and review content. Nothing else.

Read-only Auditor — **auditor** · defined, nobody holds it today

"Sees every module and all costs, changes nothing; no bank details, no salaries."

View on every module. Sees prices, sales figures and the deleted-records register. **Not one edit capability**, and no `data.view_pay`, no `data.view_banking`, no `data.view_identity`, no `data.view_reviews`.

Of the roles defined today, only this one, System Administrator and Commercial & People hold `data.view_deleted` , so only they can open Settings → Deleted records.

4.6 Reading the table above

A module capability and a data capability answer different questions. Holding `module.purchasing.view` gets you onto the purchase order list. Whether the amounts on it are figures or the word **Restricted** is decided separately, by `data.view_prices` . A role can see every order and none of the money on it.

A process can need more than one module's edit right. Raising a sales order needs `module.sales.edit` . Invoicing it needs `module.finance.edit` . Shipping needs `module.sales.edit` again but cannot happen until somebody with finance rights has invoiced. A process that crosses two modules needs two people, or one person holding both.